The Modulo-One Partition Formula

1 Setup

Fix

$$\tau \geq 2, \qquad s \geq 6, \qquad r = \tau s + 1, \qquad M = \tau\binom{s}{2},$$

and put

$$B = \tau(s-2) - 1.$$

A normalized step-τ Dyck word is a word

$$D = (x_0, \ldots, x_{s-1})$$

of nonnegative integers satisfying

$$x_0 = 0, \qquad x_{i+1} \leq x_i + \tau.$$

Its statistics are

$$\mathrm{area}(D) = \sum_i x_i, \qquad \mathrm{dinv}_\tau(D) = \sum_{i<j} d_\tau(x_i, x_j), \qquad \mathrm{defc}_\tau(D) = M - \mathrm{area}(D) - \mathrm{dinv}_\tau(D),$$

where

$$d_\tau(u,v) = \begin{cases} (u + \tau - v)_+, & u \leq v, \\ (v + \tau + 1 - u)_+, & u > v. \end{cases}$$

An occurrence $x_j = e > 0$ is extractable if exactly one earlier occurrence lies in

$$[\max(0, e-\tau), e)$$

and its deletion leaves a normalized word. A *full skeleton* is a normalized word with no extractable occurrence.

We use three root sets:

$$\begin{aligned} \mathcal{F}_B &= \{S : S \text{ is a full skeleton and } \mathrm{defc}_\tau(S) \leq B\}, \\ \mathcal{Z}_B &= \{Z : Z \text{ is normalized},\ Z_0 = Z_1 = 0,\ \mathrm{defc}_\tau(Z) \leq B\}, \\ \mathcal{P}_B^{(s-2)} &= \{\lambda : \lambda \text{ is a partition},\ \lambda_1 \leq s-2,\ |\lambda| \leq B\}. \end{aligned}$$

The empty partition is included in the last set.

We shall use the modulo-one string-decomposition theorem proved in `mod1_decomp.tex`. Thus we assume the q,t symmetry of $\mathrm{Cat}_{\tau s+1,s}(q,t)$ and the East5 and West5 nonfailure hypotheses made there. We will prove here the remaining special-skeleton area bound, rather than assume it.

2 Full skeletons and initial double zeros

Put

$$K_\tau(u,v) = \tau - d_\tau(u,v) = \begin{cases} \min(\tau, v-u), & u \le v, \\ \min(\tau, u-v-1), & u > v. \end{cases}$$

Since there are $\binom{s}{2}$ pairs, we have the useful identity

$$\operatorname{defc}_\tau(D) = \sum_{i<j} K_\tau(x_i, x_j) - \sum_j x_j. \tag{1}$$

Lemma 2.1 (Large-entry bound)**.** *Let $D = (0, 0, x_2, \ldots, x_{s-1})$ be normalized. If some entry of D is greater than τ, then*

$$\operatorname{defc}_\tau(D) \ge \tau(s-2).$$

Proof. For $1 \le j \le s-1$, form a column whose first x_j cells are 1s and whose next τ cells are 0s. A 0 and a 1 are called a contact if either the 0 is strictly to the left of the 1 in the same row, or the 1 is one row below the 0 and strictly to its left. For $i < j$, the number of contacts using columns i and j is

$$\begin{cases} \min(\tau, x_j - x_i), & x_i \le x_j, \\ \min(\tau, x_i - x_j - 1), & x_i > x_j, \end{cases}$$

which is $K_\tau(x_i, x_j)$. If $C(D)$ is the total number of contacts and

$$u(D) = \sum_{j=1}^{s-1} (x_j - \tau)_+,$$

then the pairs involving $x_0 = 0$ in (1) give

$$\operatorname{defc}_\tau(D) = C(D) - u(D). \tag{2}$$

Choose k with $x_k > \tau$. Consider the $\tau(s-2)$ cells in the first τ rows and columns $2, \ldots, s-1$. A cell containing 1 is in contact with the 0 in column 1 of the same row. If the cell is a 0 in a column left of k, it is in contact with the 1 in column k of the same row. If it is a 0 in a column right of k, it is in contact with the 1 in column k one row below. These are $\tau(s-2)$ distinct contacts.

There is also a distinct contact for each of the $u(D)$ ones below row τ. For such a 1 in row a and column j, choose $i < j$ maximal with $x_i < a$; this exists because $x_1 = 0$. Then $x_{i+1} \ge a$, and normalization gives

$$x_i < a \le x_{i+1} \le x_i + \tau.$$

Thus column i contains a 0 in row a, and it is in contact with the chosen 1 in the same row. These contacts lie below row τ, so none was counted previously. Hence

$$C(D) \ge \tau(s-2) + u(D).$$

Equation (2) proves the result. $\square$

Proposition 2.2 (The two descriptions of the roots)**.** *In the range* $\operatorname{defc}_\tau \le B$*,*

$$D \text{ is a full skeleton} \iff D \text{ begins with } (0,0).$$

Consequently $\mathcal{F}_B = \mathcal{Z}_B$.

Proof. First suppose that D is full. If $x_1 > 0$, then $x_1 \le \tau$ and $x_0 = 0$ is its unique predecessor. Since x_1 is not extractable, its outgoing splice must fail, so $x_2 > x_0 + \tau$. Normalization then makes x_2 a new record with x_1 as its unique predecessor. Repeating the same argument produces a strictly increasing record chain. The final member of this chain has a valid outgoing splice, or lies at the right boundary, and is therefore extractable, a contradiction. Hence $x_1 = 0$.

Conversely, suppose D begins with $(0,0)$ and $\mathrm{defc}_\tau(D) \le B < \tau(s-2)$. Lemma 2.1 shows that every entry is at most τ. If $e > 0$ occurs, both initial zeros lie in

$$[\max(0, e-\tau), e) = [0, e).$$

Thus e has at least two predecessors and is not extractable. Zero itself is never extractable, so D is full. □

3 Bounded partitions

We now use the one external bijection needed for the partition form of the answer.

Theorem 3.1 (Hawkes's bounded-partition bijection)**.** *There is a bijection*

$$\Phi : \mathcal{P}_B^{(s-2)} \longrightarrow \mathcal{Z}_B$$

such that

$$\mathrm{defc}_\tau(\Phi(\lambda)) = |\lambda|, \qquad \mathrm{area}(\Phi(\lambda)) = \ell(\lambda). \tag{3}$$

Source. This is the composition of the partition–matrix bijection, the matrix involution interchanging height and width, and the matrix–maximal-path bijection in Section 5 of [1]. In that paper set $\ell = s-1$ and $m = \tau$. Its maximal paths are precisely the normalized position-coordinate words beginning with $(0,0)$, and its condition $|\lambda| < (\ell-1)m$ is exactly $|\lambda| \le B$. □

Corollary 3.2 (Skeleton position)**.** *Every full skeleton S with* $\mathrm{defc}_\tau(S) \le B$ *satisfies*

$$\mathrm{area}(S) \le \mathrm{defc}_\tau(S).$$

In particular it lies strictly below the middle of its defect layer.

Proof. By Proposition 2.2 and Theorem 3.1, write $S = \Phi(\lambda)$. Since all parts of a partition are positive,

$$\mathrm{area}(S) = \ell(\lambda) \le |\lambda| = \mathrm{defc}_\tau(S).$$

The strict lower-half assertion follows from the calculation in `mod1_decomp.tex`; explicitly, if $d = \mathrm{defc}_\tau(S) \le B$, then

$$2\,\mathrm{area}(S) + d \le 3d \le 3B < M.$$

□

4 The three forms of the decomposition

Theorem 4.1 (Full-skeleton, double-zero, and partition forms)**.** *Assume the q,t symmetry and the East5 and West5 nonfailure hypotheses used in* `mod1_decomp.tex`*. The weak lower half of every defect layer $d \le B$ is partitioned into strings whose roots may be indexed equivalently by the defect-d subsets of*

$$\mathcal{F}_B, \qquad \mathcal{Z}_B, \qquad \mathcal{P}_B^{(s-2)}.$$

For the partition set, its defect-d subset means the partitions of size d. For $\lambda \in \mathcal{P}_B^{(s-2)}$, the root is $S_\lambda = \Phi(\lambda)$, and its lower-half string has one word at every area

$$\ell(\lambda),\ \ell(\lambda)+1, \ldots, \left\lfloor \frac{M-|\lambda|}{2} \right\rfloor.$$

Consequently the low-defect part of the rational q,t-Catalan polynomial has the following three equal root-indexed forms:

$$\begin{aligned}
&\sum_{\substack{D \text{ normalized} \\ \operatorname{defc}_\tau(D) \le B}} q^{\operatorname{dinv}_\tau(D)} t^{\operatorname{area}(D)} \\
&= \sum_{S \in \mathcal{F}_B} \frac{q^{\operatorname{dinv}_\tau(S)+1} t^{\operatorname{area}(S)} - q^{\operatorname{area}(S)} t^{\operatorname{dinv}_\tau(S)+1}}{q-t} && (4) \\
&= \sum_{Z \in \mathcal{Z}_B} \frac{q^{\operatorname{dinv}_\tau(Z)+1} t^{\operatorname{area}(Z)} - q^{\operatorname{area}(Z)} t^{\operatorname{dinv}_\tau(Z)+1}}{q-t} && (5) \\
&= \sum_{\lambda \in \mathcal{P}_B^{(s-2)}} \frac{q^{M-|\lambda|-\ell(\lambda)+1} t^{\ell(\lambda)} - q^{\ell(\lambda)} t^{M-|\lambda|-\ell(\lambda)+1}}{q-t}. && (6)
\end{aligned}$$

Proof. The exceptional full skeleton from `mod1_decomp.tex` has defect $\tau(s-2) = B+1$. Hence in the present range the special skeletons appearing there are exactly the full skeletons in $\mathcal{F}_B$. Corollary 3.2 proves the skeleton-position hypothesis of that decomposition theorem. Applying it gives the lower-half strings and (4).

Proposition 2.2 identifies $\mathcal{F}_B$ with $\mathcal{Z}_B$, giving (5). Finally reindex by the bijection Φ. Equation (3) and the identity

$$\operatorname{dinv}_\tau(S_\lambda) = M - \operatorname{defc}_\tau(S_\lambda) - \operatorname{area}(S_\lambda) = M - |\lambda| - \ell(\lambda)$$

give both the stated lower-half area interval and (6). $\square$

References

[1] Graham Hawkes. *A conjectured formula for the rational q,t-Catalan polynomial.* Annals of Combinatorics 28 (2024), 749–795; arXiv:2208.00577.